Teacher Wu Kuan | See It at a Glance #21 — Difference of Squares for Mental Math

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1. The trap question

99 × 101 in 3 seconds?

2. The method

• $a^2 - b^2 = (a + b)(a - b)$: cut a small square from a big one and re-glue the L-shape into a rectangle.
• $99 \times 101 = (100 - 1)(100 + 1) = 100^2 - 1^2 = 9999$.
• Spot two numbers equally far from a round number.

3. Parent check method

Ask your child to find the “middle” round number first (100) and how far each factor is from it (1).

Practice

1. $49 \times 51 = ?$
2. $98 \times 102 = ?$
3. $1003^2 - 997^2 = ?$
4. Factor $x^2 - 25$.
5. Factor $4x^2 - 9$.

Answers

1. 2499 ($50^2 - 1^2$)
2. 9996 ($100^2 - 2^2$)
3. 12000 (2000×6)
4. $(x + 5)(x - 5)$
5. $(2x + 3)(2x - 3)$

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